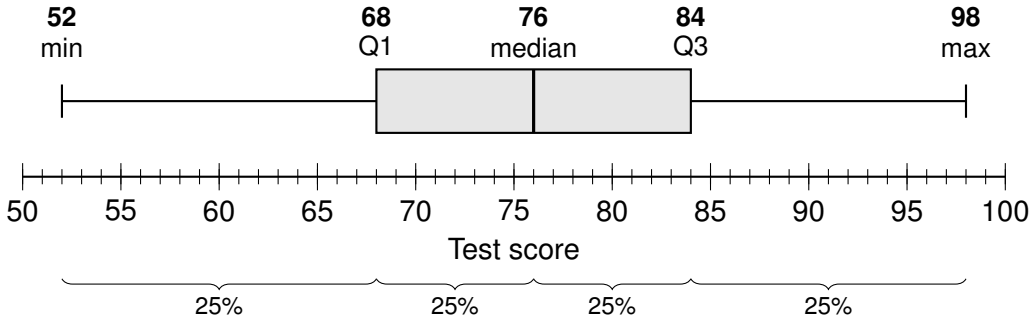
Reading Box Plots and Outliers KEY

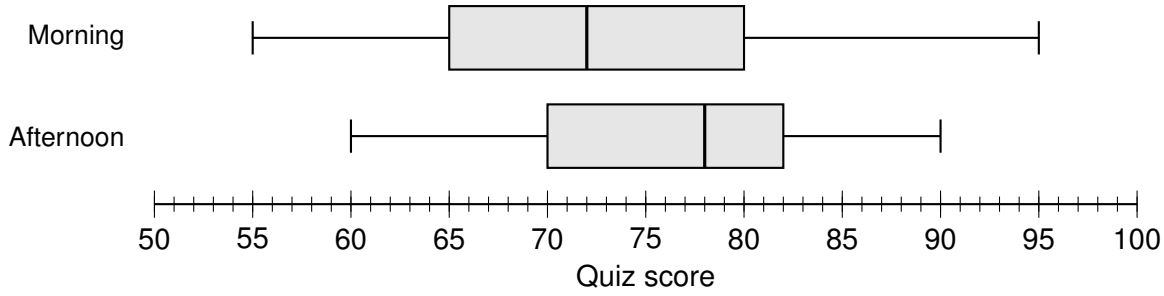
Name _____

Reading a box plot. Each of the 4 parts holds about **25%** of the data.



The box holds the middle **50%**: half of the students scored from 68 to 84.
Above Q1 is **75%**: about 75% of the students scored more than 68.

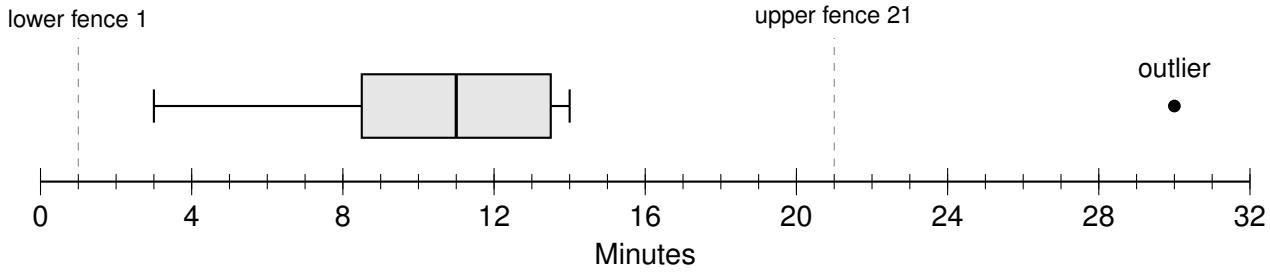
Comparing box plots. Use the same number line. Compare the **centers** and the **spreads**.



Center (median): Afternoon 78, Morning 72. The afternoon class scored higher.
Spread (IQR): Afternoon $82 - 70 = 12$, Morning $80 - 65 = 15$. Afternoon scores are closer together.

Outliers. An **outlier** is a value far away from the rest of the data.

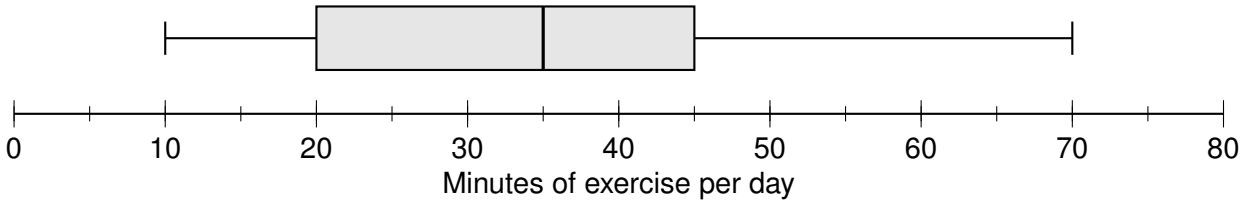
The 1.5 $\times$ IQR rule	**Example:** 3, 8, 9, 10, 11, 12, 13, 14, 30
1. Find Q1, Q3, and the IQR.	Q1 = 8.5, Q3 = 13.5, IQR = 5
2. Lower fence = Q1 $-$ 1.5 $\times$ IQR	$8.5 - 1.5 \times 5 = 8.5 - 7.5 = 1$
3. Upper fence = Q3 + 1.5 $\times$ IQR	$13.5 + 7.5 = 21$
4. A value below the lower fence or above the upper fence is an outlier.	30 is above 21, so **30 is an outlier**. 3 is not below 1, so 3 is not an outlier.



On a box plot, an outlier is a **dot**. The whisker stops at the last value that is not an outlier (here, 14).

Work with a partner. Compare answers when you finish each problem.

1. This box plot shows how many minutes some students exercise each day.



(a) Write the five-number summary. Min 10, Q1 20, Median 35, Q3 45, Max 70

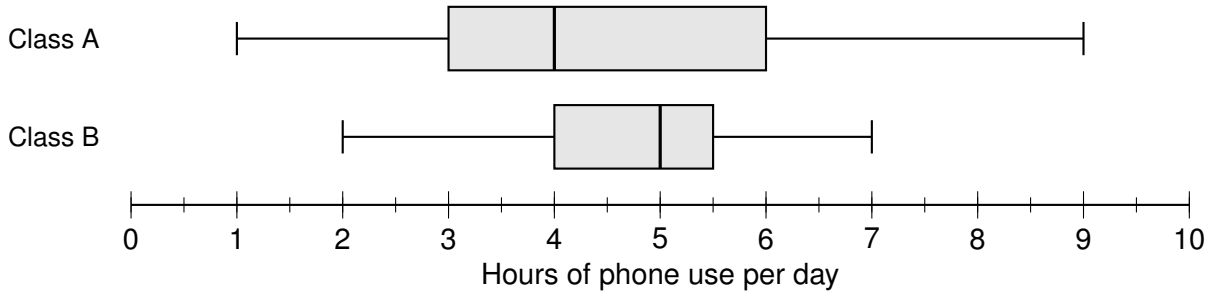
(b) Find the IQR. 45 − 20 = 25 minutes

(c) Find the range. 70 − 10 = 60 minutes

(d) About what percent of the students exercise more than 20 minutes? 75%

(e) About what percent of the students exercise from 35 to 70 minutes? 50%

2. These box plots show how many hours students in two classes use their phones each day.



(a) Which class has the higher median? Class B (5 hours vs 4 hours)

(b) Find the IQR of each class. A: 6 − 3 = 3 hours. B: 5.5 − 4 = 1.5 hours.

(c) Which class is more **consistent** (the values are closer together)? Explain.
Class B. Its IQR and its range are both smaller.

(d) About what percent of Class B uses the phone more than 4 hours a day? 75%

3. Nine students recorded how many minutes they ride the bus to school.

12, 15, 16, 18, 19, 20, 21, 22, 45

(a) Find Q1, Q3, and the IQR.
Median 19. Lower half 12, 15, 16, 18: Q1 = 15.5. Upper half 20, 21, 22, 45: Q3 = 21.5. IQR = 6.

(b) Find the lower fence and the upper fence.
$15.5 - 1.5 \times 6 = 15.5 - 9 = 6.5$. $21.5 + 9 = 30.5$

(c) Are there any outliers? Explain.
Yes. 45 is above the upper fence of 30.5. No value is below 6.5.

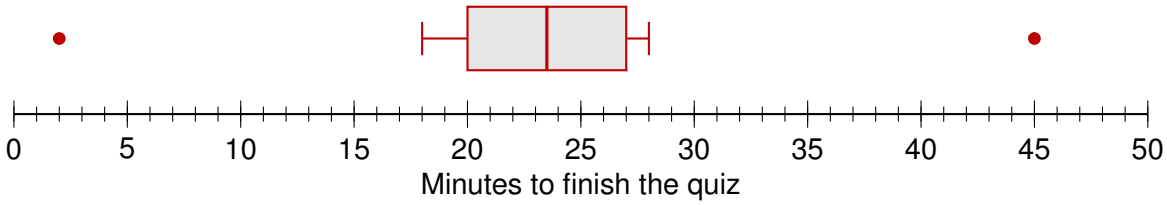
4. Ten students timed how many minutes they took to finish a quiz.

23, 45, 20, 27, 2, 24, 21, 28, 18, 25

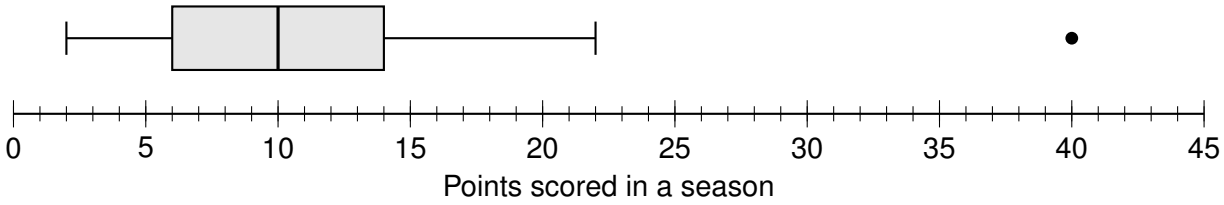
(a) Find Q1, the median, Q3, and the IQR.
2, 18, 20, 21, $\boxed{23}$, $\boxed{24}$, 25, 27, 28, 45. Median 23.5
Lower half 2, 18, 20, 21, 23: Q1 = 20. Upper half 24, 25, 27, 28, 45: Q3 = 27. IQR = 7

(b) Find the fences. Which values are outliers?
$20 - 10.5 = 9.5$ and $27 + 10.5 = 37.5$. 2 and 45 are outliers.

(c) Make a box plot. Show each outlier as a dot.



5. This box plot shows the points scored by players on a basketball team.



(a) What is the maximum? 40 points (the dot)

(b) Find the IQR. $14 - 6 = 8$ points

(c) Use the 1.5 $\times$ IQR rule to show that 40 is an outlier.
Upper fence = $14 + 1.5 \times 8 = 14 + 12 = 26$. 40 is above 26.

(d) Why does the whisker stop at 22?
22 is the largest value that is not an outlier.

6. Look at the exercise box plot in problem 1. Is each statement true or false? Explain.

(a) About half of the students exercise from 20 to 45 minutes each day.
True. The box holds the middle 50%.

(b) The right whisker is the longest part, so most students are in that part.
False. Each part has about 25% of the students. A long part means those values are spread out.

(c) The mean is 35 minutes.
False (or we cannot tell). 35 is the median. A box plot does not show the mean.

(d) At least one student exercises exactly 70 minutes.
True. The maximum is a value in the data.